

AQA A-Level Further Mathematics (7367)

Question Paper

Further Calculus

E3 - Mean of a Function

Name: _____

Class: _____

Date: _____

Time: 8 min.

Marks: 7 marks

Comments:

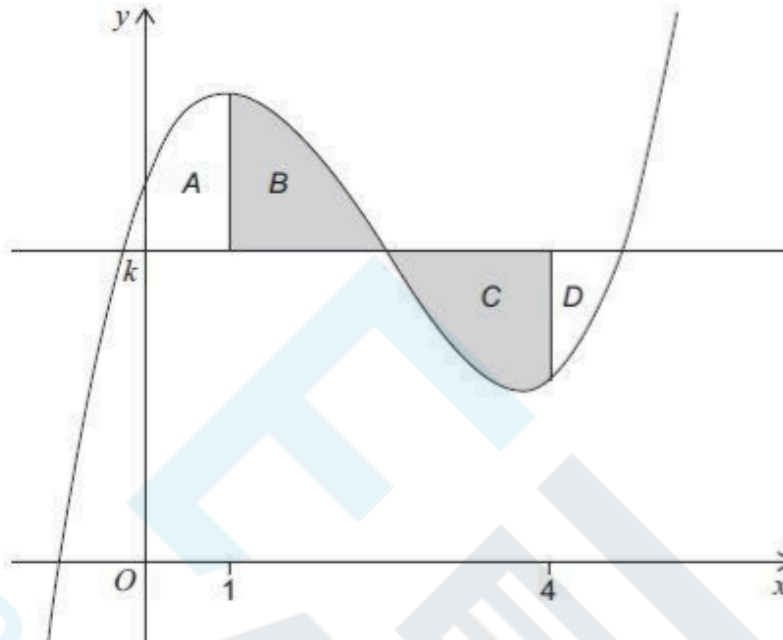
EXAM PAPERS PRACTICE
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Q1.

Four finite regions A , B , C and D are enclosed by the curve with equation

$$y = x^3 - 7x^2 + 11x + 6$$

and the lines $y = k$, $x = 1$ and $x = 4$, as shown in the diagram below.



The areas of B and C are equal.

Find the value of k .

(Total 3 marks)

Q2.

Find the mean value of $3x^2$ over the interval $1 \leq x \leq 3$

Circle your answer.

$8\frac{2}{3}$

10

13

26

(Total 1 mark)

Q3.

The function $f(x) = \cosh(ix)$ is defined over the domain $\{x \in \mathbb{R} : -a\pi \leq x \leq a\pi\}$, where a is a positive integer.

By considering the graph of $y = [f(x)]^n$, find the mean value of $[f(x)]^n$, when n is an odd positive integer.



Fully justify your answer.

(Total 3 marks)

